

An Exact Density Obstruction for Bounded Versus Precompact Sequences

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Status

This packet records a candidate partial result, likely valid, toward Problem 5.1 and Remark 5.3 of Candido [1]. It computes exactly the densities of the two sequence spaces in the problem and derives a new necessary cardinal condition for a positive example. It does not settle the surviving cases.

Source Passage

□

5. FURTHER APPLICATIONS

A natural question related to this this investigation is whether exists a Hausdorff compactum K such that

$$C(K, \ell_\infty) \sim \ell_\infty(C(K)). \quad (5.1)$$

Since $\ell_\infty \cong C(\beta\mathbb{N})$, we have $C(K, \ell_\infty) \cong C(K \times \beta\mathbb{N}) \cong C(\beta\mathbb{N}, C(K))$, and the previous question could be put into words in the following way:

Problem 5.1. *Is there a Hausdorff compactum K such that the space of bounded sequences in $C(K)$ is isomorphic to the space of precompact sequences in $C(K)$?*

As observed by D. Alspach and E. M. Galego in [1] p.157], if such infinite K exists, then $C(K)$ has a complemented subspace isomorphic to c_0 . Indeed, by [7], $c_0 \xhookrightarrow{c} C(\beta\mathbb{N}, C(K)) \sim \ell_\infty(C(K))$ and, by [20], $c_0 \xhookrightarrow{c} \ell_\infty(C(K))$ if and only if $c_0 \xhookrightarrow{c} C(K)$. We deduce, by [7] Corollary 2], that if $C(K)$ is a Grothendieck space, then $C(K, \ell_\infty) \not\sim \ell_\infty(C(K))$.

By using some of the results obtained in our investigation we can add more information.

Theorem 5.2. *Let K be an infinite Hausdorff compactum. Then*

$$C(K, \ell_\infty) \sim \ell_\infty(C(K)) \Rightarrow \ell_\infty \hookrightarrow C(K).$$

Proof. Suppose that $C(K, \ell_\infty) \sim \ell_\infty(C(K))$. Since $C(K, \ell_\infty) \sim C(\beta\mathbb{N}, C(K))$ and $C(K)$ is infinite dimensional, by Corollary 3.6, $C(\beta\mathbb{N}, C(K))$ has a complemented subspace isomorphic to $c_0(C(\beta\mathbb{N}))$. Hence $c_0(C(\beta\mathbb{N})) \xhookrightarrow{c} \ell_\infty(C(K))$. According to Corollary 4.2 there is $n \in \mathbb{N}$ such that $\ell_\infty \hookrightarrow C(K)^n$. Then, by [12], $\ell_\infty \hookrightarrow C(K)$. □

Remark 5.3. From Theorem 5.2 we deduce that $\ell_\infty(C(K)) \not\sim C(K, \ell_\infty)$ wherever K is a metric compactum. This extends the main result of [8] and [19] Corollary 3]. It would be interesting to know whether there exists an infinite Hausdorff compactum K such that (5.1) holds.

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The Question and Its Intended Scope

Candido asks whether there is a compact Hausdorff space K such that

$$C(K, \ell_\infty) \cong \ell_\infty(C(K)). \quad (1)$$

Using $\ell_\infty \cong C(\beta\mathbb{N})$ and the exponential law for continuous functions on compact spaces, the source identifies

$$C(K, \ell_\infty) \cong C(K \times \beta\mathbb{N}) \cong C(\beta\mathbb{N}, C(K)). \quad (2)$$

Restriction from $\beta\mathbb{N}$ to $\mathbb{N}$ identifies $C(\beta\mathbb{N}, C(K))$ isometrically with the bounded sequences in $C(K)$ having relatively compact range. Thus the verbal formulation of Problem 5.1 asks whether all bounded $C(K)$ -valued sequences, as a Banach space, can be isomorphic to the precompact ones.

Problem 5.1 itself does not say “infinite”, but the following paragraph, Theorem 5.2, and Remark 5.3 explicitly ask for infinite K . This restriction is essential. If K is nonempty and finite, then $C(K)$ is finite-dimensional and both sides of (1) are finite powers of ℓ_∞ , hence are isomorphic. We address the substantive infinite- K question.

Definitions and Proof Intuition

Let X be a Banach space and let Γ be a set. Write

$$\ell_\infty(\Gamma, X) = \left\{ x = (x_\gamma)_{\gamma \in \Gamma} : \sup_{\gamma \in \Gamma} \|x_\gamma\| < \infty \right\}$$

with the supremum norm, and put

$$\text{rc}_\Gamma(X) = \{x \in \ell_\infty(\Gamma, X) : \{x_\gamma : \gamma \in \Gamma\} \text{ is relatively compact in } X\}.$$

The latter is closed and is therefore a Banach space. Equivalently,

$$\text{rc}_\mathbb{N}(X) \cong C(\beta\mathbb{N}, X) \cong \mathcal{K}(\ell_1, X), \quad \ell_\infty(\mathbb{N}, X) \cong \mathcal{L}(\ell_1, X). \quad (3)$$

The operator identifications follow by sending T to $(Te_n)_{n \in \mathbb{N}}$.

The density of $\text{rc}_\Gamma(X)$ is controlled by finite-range maps, because a relatively compact set has a finite net at every scale. By contrast, $\ell_\infty(\Gamma, X)$ remembers an independent choice at every coordinate. A large separated subset of X therefore exponentiates to a large separated subset of the full bounded-family space. These observations give exact, rather than merely one-sided, density formulas.

Exact Density Formulas

Theorem 1 (Bounded versus precompact families). *Let X be an infinite-dimensional Banach space with $\text{dens } X = \kappa$, and let Γ be an infinite set with $|\Gamma| = \lambda$. Then*

$$\text{dens } \text{rc}_\Gamma(X) = \max\{\kappa, 2^\lambda\} \quad (4)$$

and

$$\text{dens } \ell_\infty(\Gamma, X) = \kappa^\lambda. \quad (5)$$

Proof. Fix a norm-dense subset $D \subset X$ of cardinality κ .

We first prove (4). Let $\mathcal{F}$ be the set of all finite-range maps from Γ to D . If $x \in \text{rc}_\Gamma(X)$ and $\varepsilon > 0$, compactness of the closure of its range gives a finite $\varepsilon/2$ -net. Approximate the points of that net by points of D and assign each coordinate one of the resulting approximants. This produces $y \in \mathcal{F}$ with $\|x - y\|_\infty < \varepsilon$. Hence $\mathcal{F}$ is dense in $\text{rc}_\Gamma(X)$.

There are κ finite subsets of D . For each fixed nontrivial finite range, the number of maps from Γ into that range is at most 2^λ . Consequently

$$|\mathcal{F}| \leq \max\{\kappa, 2^\lambda\},$$

which proves the upper bound in (4).

The constant families form an isometric copy of X in $\text{rc}_\Gamma(X)$, so its density is at least κ . Fix a unit vector $x_0 \in X$. The map

$$(a_\gamma)_{\gamma \in \Gamma} \mapsto (a_\gamma x_0)_{\gamma \in \Gamma}$$

is an isometric embedding of scalar $\ell_\infty(\Gamma)$ into $\text{rc}_\Gamma(X)$: a bounded subset of the scalar field has compact closure. Finally,

$$\text{dens } \ell_\infty(\Gamma) = 2^\lambda.$$

Indeed, the upper bound follows from cardinality, while $\{0, 1\}^\Gamma$ is a 1-separated family of cardinality 2^λ . This proves the matching lower bound in (4).

For (5), coordinatewise approximation by D shows that the bounded members of D^Γ form a dense subset of $\ell_\infty(\Gamma, X)$. Its cardinality is at most κ^λ .

For the reverse inequality, we construct a $1/2$ -separated subset S of the unit sphere of X with $|S| = \kappa$. By transfinite recursion, suppose that x_β has been chosen for $\beta < \alpha < \kappa$. The closed linear span of the earlier vectors has density at most $\max\{|\alpha|, \aleph_0\} < \kappa$, so it is a proper closed subspace of X . Riesz's lemma gives a unit vector x_α whose distance from that subspace is greater than $1/2$. The resulting family is $1/2$ -separated.

Every map from Γ into S is a bounded family, and two distinct maps have supremum distance greater than $1/2$. Thus $\ell_\infty(\Gamma, X)$ contains a uniformly separated set of cardinality

$$|S^\Gamma| = \kappa^\lambda.$$

A metric space containing an ε -separated set of cardinality μ has density at least μ . This gives the lower bound and completes the proof. $\square$

Application to $C(K)$

We use the standard equality

$$\text{dens } C(K) = w(K) \tag{6}$$

for every infinite compact Hausdorff space K . For completeness, here is a proof. If D is norm dense in $C(K)$, then D separates points: approximate a continuous function separating any prescribed pair. The evaluation map embeds K into the product of $|D|$ copies of the scalar field, so $w(K) \leq |D|$ for infinite cardinals. Conversely, a compact Hausdorff space of weight $w(K)$ has a point-separating family of continuous scalar functions of that cardinality. The unital algebra over a countable dense subfield generated by this family still has cardinality $w(K)$, and it is norm dense by the Stone–Weierstrass theorem [2, Chapter 7]. This proves (6).

Corollary 1 (Exact density test for the source problem). *Let K be an infinite compact Hausdorff space and put $\kappa = w(K)$. Then*

$$\text{dens } C(K, \ell_\infty) = \max\{\kappa, \mathfrak{c}\}, \quad \text{dens } \ell_\infty(C(K)) = \kappa^{\aleph_0}. \quad (7)$$

In particular, the two spaces have the same density if and only if

$$\kappa^{\aleph_0} = \max\{\kappa, \mathfrak{c}\}. \quad (8)$$

Proof. By (2), $C(K, \ell_\infty)$ is the precompact-range sequence space $\text{rc}_{\mathbb{N}}(C(K))$. Apply Theorem 1 with $\lambda = \aleph_0$, use $2^{\aleph_0} = \mathfrak{c}$, and invoke (6). The equality criterion is immediate. $\square$

Candido's Theorem 5.2 supplies an additional necessary condition: if (1) holds for infinite K , then ℓ_∞ embeds isomorphically into $C(K)$. Since density is monotone for subspaces of metric spaces,

$$\kappa = \text{dens } C(K) \geq \text{dens } \ell_\infty = \mathfrak{c}.$$

Combining this with Corollary 1 gives the promised obstruction.

Theorem 2 (Cardinal obstruction to a positive example). *If an infinite compact Hausdorff space K satisfies*

$$C(K, \ell_\infty) \cong \ell_\infty(C(K)),$$

then, with $\kappa = w(K)$,

$$\boxed{\kappa \geq \mathfrak{c} \quad \text{and} \quad \kappa^{\aleph_0} = \kappa.} \quad (9)$$

Consequently no such K exists whenever $w(K)^{\aleph_0} > w(K)$. In particular, no example has weight of countable cofinality.

Proof. The embedding conclusion of [1, Theorem 5.2] gives $\kappa \geq \mathfrak{c}$. Isomorphic Banach spaces have the same density, so (7) gives

$$\kappa^{\aleph_0} = \max\{\kappa, \mathfrak{c}\} = \kappa.$$

For the last assertion, suppose $\text{cf}(\kappa) = \aleph_0$. Choose an increasing sequence of cardinals $\kappa_n < \kappa$ cofinal in κ , and partition a set of cardinality κ into sets I_n with $|I_n| \leq \kappa_n$. If the product $\prod_n \kappa_n^+$ had cardinality at most κ , list all its members as $(f_\xi)_{\xi < \kappa}$. At coordinate n , fewer than κ_n^+ values occur among the $f_\xi(n)$ with $\xi \in I_n$; choose $g(n)$ outside those values. Then g differs from every f_ξ , a contradiction. Hence

$$\kappa < \prod_n \kappa_n^+ \leq \kappa^{\aleph_0},$$

so $\kappa^{\aleph_0} = \kappa$ is impossible. $\square$

Limitations and Surviving Cases

Theorem 2 is a substantial necessary condition, not a complete answer to Problem 5.1. It eliminates every weight on which countable exponentiation strictly raises cardinality. It does not distinguish the spaces when $\kappa \geq \mathfrak{c}$ and $\kappa^{\aleph_0} = \kappa$. In particular,

$$\mathfrak{c}^{\aleph_0} = (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0} = \mathfrak{c},$$

so weight $\mathfrak{c}$ survives in ZFC.

At the operator level, the remaining question asks whether, for some $X = C(K)$ satisfying the stringent structural conditions in the source,

$$\mathcal{K}(\ell_1, X) \cong \mathcal{L}(\ell_1, X).$$

The present density invariant cannot decide this fixed-point-cardinal regime. No positive example or full nonexistence proof is claimed.

Novelty Check

The run indexes were searched for the arXiv id, the exact paper title, bounded versus precompact $C(K)$ -valued sequences, both sides of (1), and the operator formulation (3). No overlapping packet was found.

Bounded external searches on 23 July 2026 used the exact phrases

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"space of bounded sequences in C(K)" "precompact sequences" isomorphic,  
"C(K,ell_infty)" "ell_infty(C(K))",  
"K(ell_1,X)" "L(ell_1,X)" isomorphic,
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as well as the paper title and citation searches. They found the arXiv and published versions of [1], but no later paper explicitly answering the problem or recording the density obstruction. This bounded search is evidence, not a guarantee, of novelty.

Human-Review Recommendation

The proof is elementary and self-contained after the source's Theorem 5.2 and the standard Stone–Weierstrass theorem. The packet should be reviewed as an exact density characterization and a new necessary condition. It should not be promoted to a full solution unless the surviving $\kappa^{\aleph_0} = \kappa$ cases are resolved by an additional Banach-space invariant.

References

- [1] L. Candido, *Complementations in $C(K, X)$ and $\ell_\infty(X)$* , Colloquium Mathematicum **172** (2023), 129–141; arXiv:2104.07152. doi:10.4064/cm8868-10-2022.
- [2] W. Rudin, *Real and Complex Analysis*, third edition, McGraw–Hill, 1987.